Plan for the Ground You Will Drive On: Forecast-Aware Off-Road Route Planning on the Vehicle

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Abstract

Off-road trafficability changes from day to day: a monsoon week can turn firm ground to mud, so a route that is passable when it is planned may not be when it is driven. Forward units also cannot count on a weather service or a network link. We forecast ground firmness one to seven days ahead on the vehicle, from a local weather log (GRU and ridge regression, 531k parameters, 12 ms per forecast on one CPU core), turn it into per-vehicle NO-GO maps with a fuzzy AND of slope, land cover and soil, and plan for the day of travel. A convoy shares only its tracks, so each vehicle accounts for ground churned by those ahead. In a backtest of 325,800 trips over two monsoons on the Pachmarhi plateau, planning three days ahead on the forecast sent 24% fewer trips into mud than planning on today's map, and at equal risk held back 31–40% fewer trips than a safety margin on today's map.

CCS Concepts

• **Computer systems organization** → **Embedded and cyber-physical systems**; • **Computing methodologies** → *Neural networks.*

Keywords

edge computing, trafficability, route planning, soil strength forecasting, convoy coordination

1 Problem

Off the road, the soil decides which vehicles move. The NATO Reference Mobility Model rates soil strength as a rating cone index (RCI) and each vehicle by a vehicle cone index: a vehicle is stuck on the first pass where $\mathrm{RCI} < \mathrm{VCI}_1$ and can sustain traffic where $\mathrm{RCI} \geq \mathrm{VCI}_{50}$ [7]. Dynamic trafficability maps combine terrain and weather factors for the time they are built [4]. Two things are missing. The ground on the day of travel differs from the day of planning, and forward units planning off the network have no weather service to ask. We test whether a forecast small enough to run on the vehicle produces better routes, and what a convoy must exchange to plan together.

2 System

Ground firmness. The site is the Pachmarhi plateau (Satpura range, 1,067 m), with a daily record for 2015–2025 (4,018 days): ERA5-Land soil moisture in two layers [5] and IMD rainfall (ERA5 for 2025). With each layer scaled to its 11-year range, $\mathrm{SWI} = 0.35\, s_{0-7} + 0.65\, s_{7-28}$ and $\mathrm{RCI} = 300\, e^{-3.0\,\mathrm{SWI}}\,(1 - 0.15 \min(R_{3d}/50\,\mathrm{mm}, 1))$ psi, where R_{3d} is the last three days' rain. The saturated value, 15 psi, follows field readings on lanes where a 4×4 bogged down [7].

Vehicles. From a public 2,451-row vehicle-attribute list we keep 1,145 ground vehicles (1,004 tracked, 125 wheeled, 16 half-tracks). Each gets VCI_1 and VCI_{50} from the Mobility Index [3, 7], with running gear assumed by weight class. Our tracked implementation reproduces the published M1 example (MI 98.6, VCI_1 26.3). Results are reported for five VCI_1 bands, which are NO-GO on 10–30% of days.

Edge forecaster. A GRU (128×2) [2] and a ridge regression, averaged, predict the change in RCI 1–7 days ahead from 60 days of local history, with no numerical weather prediction. A safety margin per band is subtracted before the comparison with VCI_1. Each margin is tuned to catch 90% of NO-GO days on one test year and scored on the other.

Maps and planning. The plateau-wide RCI is spread over a 90 m grid (76k cells) by topographic wetness index [1], shifting each cell's SWI by up to ±0.2. Terrain comes from a Google Earth Engine export (DEM, ESA WorldCover [9], Sentinel-2 NDVI as canopy density) and 403 OpenStreetMap road segments [6]. Slope and land-cover memberships follow Liu et al.'s tables [4]. Soil membership is 0 below VCI_1 and rises to 1 at VCI_{50}, so SLOW-GO is slow, not forbidden. Cells combine by fuzzy AND (minimum) [8], which vetoes deal-breakers by construction. Routes minimise travel time with speed proportional to membership (Dijkstra, equivalent to A*).

Convoys. Each pass raises the firmness the next vehicle needs, which we model as $\mathrm{VCI}(n) = \mathrm{VCI}_1 + (\mathrm{VCI}_{50} - \mathrm{VCI}_1) \ln n / \ln 50$. In the wear-aware mode each vehicle broadcasts its track (4 B per cell), and the next one plans on the shared pass-count map.

3 Evaluation

Forecast. Over the 2024–25 test years, the 7-day RMSE is 20.8 psi against 26.0 for persistence (20% lower). At 90% NO-GO recall the forecast makes 23–45% fewer false NO-GO calls per vehicle-year than the same margin rule applied to today's value.

Routing backtest. For every test day, band and lead time $h \in \{1, 3, 7\}$ we plan 30 trips of 8–25 km between points on the road network and score each plan against the real ground on the day of travel. A plan is *stuck* if its route crosses a NO-GO cell and *held* if it finds no route when one existed (Table 1). Planning on the forecast beats today's map on both errors: 1.6% against 2.1% stuck at 3 days (−24%) and 2.9% against 3.9% at 7 days (−26%), with fewer trips held back. A margin on today's value is slightly safer at its tuned setting but holds back far more trips. Sweeping both margins (Fig. 1) shows the forecast dominating: at an equal stuck rate it holds back 40% (3 days) and 31% (7 days) fewer trips. Extra travel time against the best route is under 0.1%. A weighted mean of factor scores without a veto sends 34.8% of trips into mud; with Liu

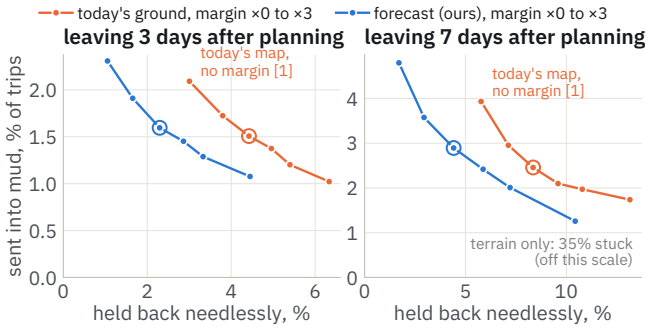


Figure 1: Both planning errors as each planner's safety margin goes from none to 3× the tuned value (ringed). Lower left is better.

Table 1: Routing backtest, 2024–25: share of trips sent into mud (stuck) and of possible trips held back, by planner and lead time.

	3 days		7 days	
Planner	stuck	held	stuck	held
Terrain only	35.5%	0.0%	35.5%	0.0%
Today's map [4]	2.1%	3.0%	3.9%	5.8%
Today + margin	1.5%	4.4%	2.5%	8.4%
Forecast + margin	**1.6%**	**2.3%**	**2.9%**	**4.4%**

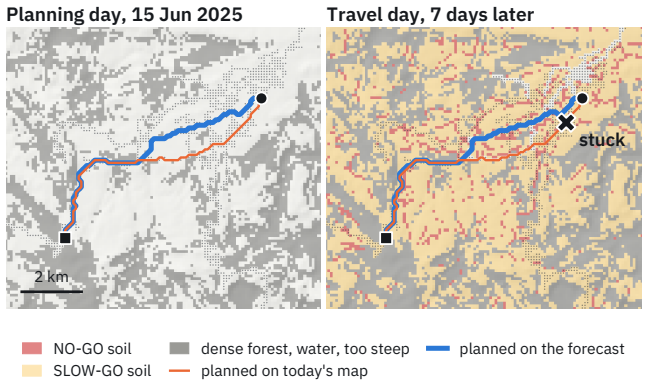


Figure 2: One trip from the backtest. Planned on 15 June 2025 for 7 days later, the route on today's map (orange) crosses ground that is NO-GO by the travel day (×); the route planned on the forecast (blue) avoids it.

et al.'s veto, 1.8%. No model foresees the day the monsoon breaks without a weather forecast; the gain comes in the days around the transitions.

Convoys and the network. On 54 soft days, ten heavy tracked vehicles that follow the leader get 5.5 through on average; re-planning on the shared pass-count map gets 9.2 through, for 6.9 kB of track updates per convoy. Wheeled vehicles gain less (6.4 to 7.0) because the forest leaves them few parallel lanes. A 7-day forecast is 18 B, against 76 kB for one passability map, so the link carries 4,200× less when vehicles rebuild maps locally. The model is 2.0 MB, and re-planning a route takes 25 ms.

4 Limitations and Next Steps

RCI is computed from modelled soil moisture in one 9 km cell, not measured, and spreading it by wetness index has no field check. Vehicle masses come from a game-file dataset, with wheeled masses corrected from open sources, so we report bands rather than named vehicles. Tuned margins differ between the two test years by up to 7.2 psi. Next: cone-penetrometer readings at Pachmarhi, Sentinel-1 radar as a soil-moisture input, persistent ruts, and convoy trials over real tactical radio links.

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